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Math 208 Final Exam Term 231

Original exam date: 19 December 2023. Practice version: questions and choices shuffled.
Net time allowed: 180 minutes

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1. The minimum radius of convergence of the power series solutions for the differential equation $(x^2 - 2x + 5)y'' + xy' - y = 0$ about the ordinary point $x = 0$ is

(a) 2

(b) 0

(c) $\sqrt{5}$

(d) ∞

(e) $\sqrt{8}$

2. Let $F(t) = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$, $e^{At} = \begin{bmatrix} e^t & 0 \\ 0 & e^{2t} \end{bmatrix}$. A particular solution for the system $X' = AX + F(t)$ is

(a) $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$

(b) $\begin{bmatrix} -3 \\ 2 \end{bmatrix}$

(c) $\begin{bmatrix} -3 \\ 1 \end{bmatrix}$

(d) $\begin{bmatrix} 3 \\ \frac{1}{2} \end{bmatrix}$

(e) $\begin{bmatrix} -3 \\ \frac{1}{2} \end{bmatrix}$

3. The solution of the homogeneous differential equation $\dfrac{dy}{dx} = \dfrac{y - x}{y + x}$ is given by

(a) $\ln(x^2 + y^2) + 5\tan^{-1}\left(\dfrac{y}{x}\right) = c$

(b) $\ln(x^2 + y^2) + 3\tan^{-1}\left(\dfrac{y}{x}\right) = c$

(c) $\ln(x^2 + y^2) - 4\tan^{-1}\left(\dfrac{y}{x}\right) = c$

(d) $\ln(x^2 + y^2) - \tan^{-1}\left(\dfrac{y}{x}\right) = c$

(e) $\ln(x^2 + y^2) + 2\tan^{-1}\left(\dfrac{y}{x}\right) = c$

4. The general solution of the system $X' = \begin{bmatrix} 4 & 1 \\ 6 & -1 \end{bmatrix} X$ is given by

(a) $X = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{5t} + c_2 \begin{bmatrix} 0 \\ 2 \end{bmatrix} e^{-2t}$

(b) $X = c_1 \begin{bmatrix} 1 \\ -6 \end{bmatrix} e^{-2t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{5t}$

(c) $X = c_1 \begin{bmatrix} 1 \\ 6 \end{bmatrix} e^{5t} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{-2t}$

(d) $X = c_1 \begin{bmatrix} 1 \\ -6 \end{bmatrix} e^{5t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-2t}$

(e) $X = c_1 \begin{bmatrix} 1 \\ 6 \end{bmatrix} e^{-2t} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{5t}$

5. Let $A = \begin{bmatrix} 5 & -3 \\ 2 & 0 \end{bmatrix}$. If P is diagonalizing matrix such that $P^{-1}AP = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, then

(a) $P = \begin{bmatrix} 1 & 3 \\ -1 & 2 \end{bmatrix}$

(b) $P = \begin{bmatrix} 1 & 3 \\ 1 & -2 \end{bmatrix}$

(c) $P = \begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix}$

(d) $P = \begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix}$

(e) $P = \begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix}$

6. An appropriate form of a particular solution of the differential equation $y^{(3)} - y'' - 12y' = x - 2xe^{-3x}$ is

(a) $y_p = A + Bx + (Cx + Dx^2)e^{-3x}$

(b) $y_p = Ax + Bx^3 + (Cx + Dx^2)e^{-3x}$

(c) $y_p = Ax + Bx^2 + (Cx^2 + Dx^3)e^{-3x}$

(d) $y_p = Ax + Bx^2 + (Cx + Dx^2)e^{-3x}$

(e) $y_p = Ax + Bx^2 + (C + Dx^2)e^{-3x}$

7. The solution of the exact differential equation $(2xy^2 - 3)\,dx + (2x^2y + 4)\,dy = 0$ is given by

(a) $x^2y^2 - 3x + 4y = c$

(b) $x^2y^3 - 3x - 4y = c$

(c) $x^2y^2 + 3x - 3y = c$

(d) $x^2y^3 + 3x + 4y = c$

(e) $x^2y^2 + 4x + 3y = c$

8. The roots of the indicial equation at $x = 0$ for the differential equation $2xy'' - y' + 2y = 0$ are

(a) $r_1 = 0,\ r_2 = -\dfrac{5}{2}$

(b) $r_1 = 1,\ r_2 = \dfrac{5}{2}$

(c) $r_1 = 0,\ r_2 = \dfrac{5}{2}$

(d) $r_1 = 1,\ r_2 = \dfrac{3}{2}$

(e) $r_1 = 0,\ r_2 = \dfrac{3}{2}$

9. If $V = \begin{bmatrix} \alpha \\ 1 \\ \beta \end{bmatrix}$ is an eigenvector of the matrix $A = \begin{bmatrix} 4 & -3 & 1 \\ 2 & -1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ associated with the eigenvalue $\lambda = 1$, then $10\alpha + 6\beta =$

(a) 4

(b) 10

(c) 6

(d) 0

(e) 8

10. If $y(x)$ is the solution of the initial-value problem $y'' + 4y = 2x,\ y(0) = 1,\ y'(0) = 2$, then $y(2\pi) =$

(a) π

(b) $2 - \pi$

(c) $2 + \pi$

(d) $1 - \pi$

(e) $1 + \pi$

11. The solution of the linear differential equation $x\dfrac{dy}{dx} - y = x^2 \sin x$ is given by

(a) $y = cx + x\sin x$

(b) $y = cx - x\cos x$

(c) $y = cx - x^2\cos x$

(d) $y = cx^2 + x\cos x$

(e) $y = cx^2 + x\sin x$

12. The general solution of the differential equation $(D+1)(D-2)^2(D^2-4D+13)y = 0$ is given by

(a) $y = c_1e^{-x} + c_2e^{2x} + c_3e^{2x}\cos(3x) + c_4e^{2x}\sin(3x)$

(b) $y = c_1e^{-x} + c_2e^{2x} + c_3xe^{2x} + c_4e^{3x}\cos(2x) + c_5e^{3x}\sin(3x)$

(c) $y = c_1e^{-x} + c_2e^{2x} + c_3xe^{2x} + c_4e^{2x}\cos(3x) + c_5e^{2x}\sin(3x)$

(d) $y = c_1e^{x} + c_2e^{2x} + c_3xe^{2x} + c_4e^{3x}\cos(2x) + c_5e^{3x}\sin(2x)$

(e) $y = c_1e^{-x} + c_2e^{-2x} + c_3xe^{-2x} + c_4e^{2x}\cos(3x) + c_5e^{2x}\sin(3x)$

13. Let $A = \begin{bmatrix} 6 & 4 \\ -9 & -6 \end{bmatrix}$, then $e^{At} =$

(a) $\begin{bmatrix} 1+6t & -4t \\ -9t & 1-6t \end{bmatrix}$

(b) $\begin{bmatrix} 1+6t & -4t \\ 9t & 1+6t \end{bmatrix}$

(c) $\begin{bmatrix} 1+6t & 4t \\ 9t & 1+6t \end{bmatrix}$

(d) $\begin{bmatrix} 1+6t & 4t \\ -9t & 1-6t \end{bmatrix}$

(e) $\begin{bmatrix} 1-6t & 4t \\ 9t & 1-6t \end{bmatrix}$

14. If $y(x)$ is the solution of the initial value problem $e^y\,dx - e^{-x}\,dy = 0,\ y(0) = 0$, then $y(x) =$

(a) $y = \ln(2 - e^x)$

(b) $y = 2\ln(2 - e^x)$

(c) $y = -3\ln(2 - e^x)$

(d) $y = -\ln(2 - e^x)$

(e) $y = 3\ln(2 - e^x)$

15. A possible fundamental matrix for the system $X' = \begin{bmatrix} 2 & -1 \\ -4 & 2 \end{bmatrix} X$ is

(a) $\Phi(t) = \begin{bmatrix} -1 & e^{4t} \\ 2 & -2e^{4t} \end{bmatrix}$

(b) $\Phi(t) = \begin{bmatrix} 1 & -e^{4t} \\ 2 & 3e^{4t} \end{bmatrix}$

(c) $\Phi(t) = \begin{bmatrix} 1 & e^{4t} \\ -2 & -2e^{4t} \end{bmatrix}$

(d) $\Phi(t) = \begin{bmatrix} 1 & e^{4t} \\ 2 & 2e^{4t} \end{bmatrix}$

(e) $\Phi(t) = \begin{bmatrix} 1 & e^{4t} \\ 2 & -2e^{4t} \end{bmatrix}$

16. If the general solution of the system $X' = \begin{bmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{bmatrix} X$ is given by $X = c_1 \begin{bmatrix} a \\ b \\ -2 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} \alpha \\ \beta \\ 1 \end{bmatrix} e^{\lambda t} + c_3 \begin{bmatrix} e \\ f \\ 13 \end{bmatrix}$, then $a \cdot b \cdot \lambda =$

(a) 12

(b) -24

(c) 24

(d) -12

(e) 0

17. Using variation of parameters, the differential equation $y'' - 2y' + y = \dfrac{e^x}{1+x^2}$ has a particular solution $y_p = u_1(x)e^x + u_2(x)xe^x$, then $u_1(x) =$

(a) $-3\ln(1+x^2)$

(b) $-\dfrac{1}{2}\ln(1+x^2)$

(c) $\dfrac{3}{2}\ln(1+x^2)$

(d) $\ln(1+x^2)$

(e) $3\ln(1+x^2)$

18. A basis for the subspace $W = \{(x, y, z) : x - 2y + 5z = 0\}$ of $\mathbb{R}^3$ consist of

(a) $v_1 = (2, 1, 0)$ and $v_2 = (-5, 0, 1)$

(b) $v_1 = (2, 1, 0)$ and $v_2 = (1, 0, 1)$

(c) $v_1 = (2, 1, 0)$ and $v_2 = (-5, 1, 1)$

(d) $v_1 = (2, 1, 1)$ and $v_2 = (-5, 0, 1)$

(e) $v_1 = (0, 1, 0)$ and $v_2 = (-5, 0, 1)$

19. An eigenvector associated with eigenvalue $\lambda = 2i$ of the matrix $A = \begin{bmatrix} 2 & 8 \\ -1 & -2 \end{bmatrix}$ is

(a) $\begin{bmatrix} 2+2i \\ -4 \end{bmatrix}$

(b) $\begin{bmatrix} 2+2i \\ -1 \end{bmatrix}$

(c) $\begin{bmatrix} 2+2i \\ -3 \end{bmatrix}$

(d) $\begin{bmatrix} 2+2i \\ -5 \end{bmatrix}$

(e) $\begin{bmatrix} 2+2i \\ -2 \end{bmatrix}$

20. Let $t = (2, -7, 9)$, $u = (1, -2, 2)$, $v = (3, 0, 1)$, $w = (1, -1, 2)$ be four vectors in $\mathbb{R}^3$. If $t = au + bv + cw$, then $a^2 + b^2 + c^2 =$

(a) 16

(b) 14

(c) 18

(d) 17

(e) 15

21. The largest eigenvalue of the matrix $A = \begin{bmatrix} 4 & -3 & 1 \\ 2 & -1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ is equal to

(a) 2

(b) -4

(c) 3

(d) 5

(e) 1

22. If $X = c_1 \begin{bmatrix} 5 \\ -6 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{4t}$ is the solution of the initial value problem $X' = \begin{bmatrix} 9 & 5 \\ -6 & -2 \end{bmatrix} X$, $X(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, then $c_2 - c_1 =$

(a) 6

(b) 3

(c) 4

(d) 8

(e) 7

23. The matrix $A = \begin{bmatrix} 5 & 1 & 6 \\ -1 & 1 & -3 \\ 0 & 1 & 3 \end{bmatrix}$ has a defective eigenvalue $\lambda = 3$ of defect 2. Choosing $V_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ such that $(A - 3I)^3 V_3 = 0$ and $(A - 3I)^2 V_3 \neq 0$, then the general solution of the system $X' = AX$ is

(a) $X = \left(c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2+3t \\ 1 \\ t \end{bmatrix} + c_3 \begin{bmatrix} 1+t+\frac{t^2}{2} \\ t \\ 1 \end{bmatrix} \right) e^{3t}$

(b) $X = \left(c_1 \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} 2+3t \\ -1 \\ -t \end{bmatrix} + c_3 \begin{bmatrix} 1+2t+\frac{3t^2}{2} \\ -t \\ -\frac{t^2}{2} \end{bmatrix} \right) e^{3t}$

(c) $X = \left(c_1 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} 3t \\ 1 \\ t \end{bmatrix} + c_3 \begin{bmatrix} 1+2t+\frac{3t^2}{2} \\ -t \\ -\frac{t^2}{2} \end{bmatrix} \right) e^{3t}$

(d) $X = \left(c_1 \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} 3t \\ 1 \\ t \end{bmatrix} + c_3 \begin{bmatrix} \frac{t^2}{2} \\ t \\ t^2 \end{bmatrix} \right) e^{3t}$

(e) $X = \left(c_1 \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} 3t \\ -1 \\ -t \end{bmatrix} + c_3 \begin{bmatrix} 1+2t+\frac{t^2}{2} \\ -t \\ -\frac{t^2}{2} \end{bmatrix} \right) e^{3t}$

24. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 7 & 2 \\ 10 & -15 & -4 \end{bmatrix}$. A basis for the eigenspace of A associated with the eigenvalue $\lambda = 1$ of A is

(a) $\left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \right\}$

(b) $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \right\}$

(c) $\left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \right\}$

(d) $\left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -3 \end{bmatrix} \right\}$

(e) $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} \right\}$

25. Let $A = \begin{bmatrix} 5 & -4 \\ 3 & -2 \end{bmatrix}$, $\Phi(t) = \begin{bmatrix} e^t & 4e^{2t} \\ e^t & 3e^{2t} \end{bmatrix}$. If $\Phi(t)$ is a fundamental matrix for the system $X' = AX$, then $e^{At} =$

(a) $\begin{bmatrix} -3e^t + 4e^{2t} & e^t - e^{2t} \\ -3e^t + 3e^{2t} & 4e^t - 3e^{2t} \end{bmatrix}$

(b) $\begin{bmatrix} 3e^t - 4e^{2t} & 4e^t - 4e^{2t} \\ -3e^t + 3e^{2t} & 4e^t - 3e^{2t} \end{bmatrix}$

(c) $\begin{bmatrix} -3e^t + 4e^{2t} & 4e^t - 4e^{2t} \\ e^t + 3e^{2t} & 4e^t - e^{2t} \end{bmatrix}$

(d) $\begin{bmatrix} -3e^t + 4e^{2t} & 5e^t - 5e^{2t} \\ -2e^t + 2e^{2t} & 4e^t - 3e^{2t} \end{bmatrix}$

(e) $\begin{bmatrix} -3e^t + 4e^{2t} & 4e^t - 4e^{2t} \\ -3e^t + 3e^{2t} & 4e^t - 3e^{2t} \end{bmatrix}$

26. If $y = \sum_{n=0}^{\infty} C_n x^n$ is a power series solution about the ordinary point $x = 0$ of the differential equation $y'' - 2xy' + y = 0$, then the coefficients C_n satisfy

(a) $C_{n+2} = \dfrac{2}{(n+2)(n+1)} C_{n-1},\ n \geq 1$

(b) $C_{n+2} = \dfrac{2n-1}{(n+2)(n+1)} C_n,\ n \geq 1$

(c) $C_{n+2} = \dfrac{2n+1}{(n+2)(n+1)} C_n,\ n \geq 1$

(d) $C_{n+2} = \dfrac{2n}{(n+1)(n+2)} C_n,\ n \geq 1$

(e) $C_{n+2} = \dfrac{3}{(n+1)(n+2)} C_{n-2},\ n \geq 1$

27. The rank of the matrix $A = \begin{bmatrix} 1 & -3 & 0 & -5 \\ -1 & 4 & 1 & 7 \\ 2 & 1 & 7 & 4 \\ 2 & -2 & 4 & -2 \end{bmatrix}$ is equal to

(a) 4

(b) 2

(c) 1

(d) 0

(e) 3

28. A 2×2 real matrix A has an eigenvector $\begin{bmatrix} 2+2i \\ -1 \end{bmatrix}$ associated with the eigenvalue $\lambda = 2i$ of A. Then the general solution of the system $X' = AX$ is

(a) $X = c_1 \begin{bmatrix} \cos(2t) - 2\sin(2t) \\ -\cos(2t) \end{bmatrix} + c_2 \begin{bmatrix} \cos(2t) + 2\sin(2t) \\ -\sin(2t) \end{bmatrix}$

(b) $X = c_1 \begin{bmatrix} 2\cos(2t) - 2\sin(2t) \\ -2\cos(2t) \end{bmatrix} + c_2 \begin{bmatrix} 2\cos(2t) + 2\sin(2t) \\ -\sin(2t) \end{bmatrix}$

(c) $X = c_1 \begin{bmatrix} 2\cos(2t) - 2\sin(2t) \\ -2\cos(2t) \end{bmatrix} + c_2 \begin{bmatrix} 2\cos(2t) + 2\sin(2t) \\ -2\sin(2t) \end{bmatrix}$

(d) $X = c_1 \begin{bmatrix} 2\cos(2t) - 2\sin(2t) \\ -\cos(2t) \end{bmatrix} + c_2 \begin{bmatrix} 2\cos(2t) + 2\sin(2t) \\ -\sin(2t) \end{bmatrix}$

(e) $X = c_1 \begin{bmatrix} 2\cos(2t) - \sin(2t) \\ -\cos(2t) \end{bmatrix} + c_2 \begin{bmatrix} 2\cos(2t) + \sin(2t) \\ -\sin(2t) \end{bmatrix}$

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	Zeros $1 \pm 2i$ are at distance $\sqrt{5}$ from 0.
2	**(e)**	$X_p = e^{At} \int e^{-At} F\, dt = (-3,\ \frac{1}{2})^T$.
3	**(e)**	$v = y/x$: $\frac{v+1}{1+v^2} dv = -\frac{dx}{x}$, so $\ln(x^2+y^2) + 2\tan^{-1}\frac{y}{x} = c$.
4	**(b)**	$\lambda = -2$: $(1,-6)$; $\lambda = 5$: $(1,1)$.
5	**(e)**	$\lambda = 2$: $(1,1)$; $\lambda = 3$: $(3,2)$; $P = [V_1\ V_2]$.
6	**(d)**	Roots $0, 4, -3$: both groups overlap, multiply each by x.
7	**(a)**	$F = x^2y^2 - 3x + g(y)$, $g' = 4$: $x^2y^2 - 3x + 4y = c$.
8	**(e)**	$r(r-1) - \frac{1}{2}r = 0$, so $r = 0,\ \frac{3}{2}$.
9	**(b)**	$(A - I)V = 0$: $\beta = 0$, $\alpha = 1$, so $10\alpha + 6\beta = 10$.
10	**(e)**	$y = \cos 2x + \frac{3}{4}\sin 2x + \frac{x}{2}$, so $y(2\pi) = 1 + \pi$.
11	**(b)**	$\rho = 1/x$: $(y/x)' = \sin x$, so $y = cx - x\cos x$.
12	**(c)**	Roots $-1,\ 2,\ 2,\ 2 \pm 3i$ give the five independent solutions listed.
13	**(d)**	$A^2 = 0$, so $e^{At} = I + At$.
14	**(d)**	Separable: $e^x = -e^{-y} + C$, $C = 2$, so $y = -\ln(2 - e^x)$.

Q	Ans	Brief explanation
15	**(e)**	$\lambda = 0$: $(1, 2)$; $\lambda = 4$: $(1, -2)$; columns $Ve^{\lambda t}$.
16	**(b)**	Eigenvalues $3, 0, -4$; $(a, b) = (2, 3)$, $\lambda = -4$, so $ab\lambda = -24$.
17	**(b)**	$W = e^{2x}$, $u_1' = -\frac{x}{1+x^2}$, so $u_1 = -\frac{1}{2}\ln(1 + x^2)$.
18	**(a)**	$x = 2y - 5z$: $(x, y, z) = y(2, 1, 0) + z(-5, 0, 1)$.
19	**(b)**	$(2 - 2i)z_1 + 8z_2 = 0$ with $z_2 = -1$ gives $z_1 = 2 + 2i$.
20	**(b)**	Solve: $a = 2,\ b = -1,\ c = 3$, so $a^2 + b^2 + c^2 = 14$.
21	**(a)**	$\det(A - \lambda I) = (2 - \lambda)(\lambda - 1)(\lambda - 2)$, largest eigenvalue 2.
22	**(e)**	$5c_1 + c_2 = 1$, $6c_1 + c_2 = 0$: $c_1 = -1$, $c_2 = 6$, so $c_2 - c_1 = 7$.
23	**(b)**	Chain $(3, 0, -1) \leftarrow (2, -1, 0) \leftarrow (1, 0, 0)$ gives the three solutions.
24	**(d)**	$A - I$ has rank 1: $z = 2x - 3y$, basis $(1, 0, 2), (0, 1, -3)$.
25	**(e)**	$e^{At} = \Phi(t)\Phi(0)^{-1}$, $\Phi(0)^{-1} = \begin{bmatrix} -3 & 4 \\ 1 & -1 \end{bmatrix}$.
26	**(b)**	$(n + 2)(n + 1)C_{n+2} = (2n - 1)C_n$.
27	**(b)**	Row reduce: rows 3 and 4 become zero, leaving 2 pivots, so rank = 2.
28	**(d)**	Real and imaginary parts of Ve^{2it} give the two solutions.